Monthly Intelligence Report

Edition 002 — Mexico

Where hurricane resilience and seismic preparedness challenge grid stability. edition · May 2026 · SSI v4.0.2

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SECTION A

Executive Summary

SUBSTATIONS SCORED

3,140

HV + MV across 32 Estados

GRID LINES MAPPED

~30,396

~30,000 km transmission & distribution

MC SIMULATIONS

~31M

10,000 per substation

PUBLIC DATA SOURCES

30+

95 variables · zero proprietary

The SSI Index is the only operational framework that fuses substation-level engineering metrics with socio-economic consequence valuation in a single composite score — scoring 71/80 (89%) across 16 competitive dimensions. Its engine processes ~52,000 source data points per run, executes ~31 million Monte Carlo simulations through a 20×20 Gaussian copula correlation matrix, and performs ~400 million arithmetic operations — producing ~232,000 output data points with full uncertainty quantification across 3,140 substations and ~30,396 grid lines spanning ~30,000 km.

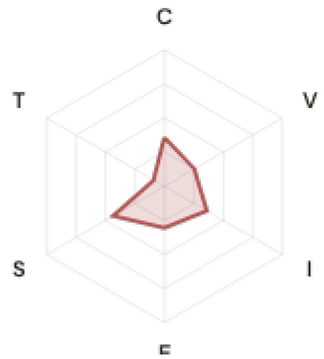
Operated as a non-profit through a dedicated foundation, the SSI Index serves the public interest with zero dependency on proprietary data. All 95 variables are drawn from 30+ verified public sources (CENACE, CRE, CFE, INEGI, SMN, CENAPRED), making the methodology fully auditable and reproducible. Initially covering Mexico's CENACE-managed transmission and ~250 CFE distribution divisions (DSOs), the Index is progressively being rolled out to all OECD countries — applying the same silo-breaking approach: fusing grid risk with societal exposure at asset-level resolution wherever open data permits.

Substation in Focus

Subestación Servicios La Yesca

Jalisco · Jalisco · 115 kV

0.433	Medium	0.152	77th
R_FINAL	CLASSIFICATION	CI WIDTH	FLEET PERCENTILE



COMPONENTS

- **C** Continuity: 0.35 / 0.30 (36%)
- **I** Infrastructure: 0.36 / 0.25 (36%)
- **S** Saturation: 0.44 / 0.20 (44%)
- **V** Voltage: 0.25 / 0.10 (25%)
- **E** Economic: 0.30 / 0.10 (30%)
- **T** Transition: 0.09 / 0.05 (9%)

MODIFIERS

- **R3 Consequence + Socio-Economic:** 1.042 ↑ amplifies
- **R4 Graph Criticality:** 1.019 ↑ amplifies
- **R6a Restoration:** 1.012 ↑ amplifies
- **R6b Hurricane + Seismic + Volcanic:** 1.169 ↑ amplifies
- **R7 Digital Readiness:** 1.006 → neutral

This month's spotlight — automatically selected for May 2026 — is **Subestación Servicios La Yesca** in Jalisco, Jalisco. With an R_final of 0.433 (Medium band, 77th fleet percentile), its risk profile is dominated by the E (Economic) component at 304% of its weight ceiling, followed by V (Voltage) at 252%. Modifiers collectively shift R_base by 26.0%, reflecting the substation's specific geographic, socio-economic, and network context. The component profile illustrates the SSI's core principle: risk is never mono-dimensional. Even when one component dominates, the interaction of all six — modulated by the five multipliers — determines the final score that drives investment prioritisation.

SECTION B — RECURRING MODULE

Cyber & Economic Exposure Monitor

Why this section exists: Traditional grid risk frameworks treat cybersecurity and economic vulnerability as separate domains. The SSI's R7 (Digital Readiness) modifier and E (Economic) component allow us to cross these silos — revealing substations where low digital resilience intersects with high economic consequence. This is precisely the anticipatory intelligence that Mexico's Energy Regulatory Commission (CRE) and CENACE network operator strategic planning requires.

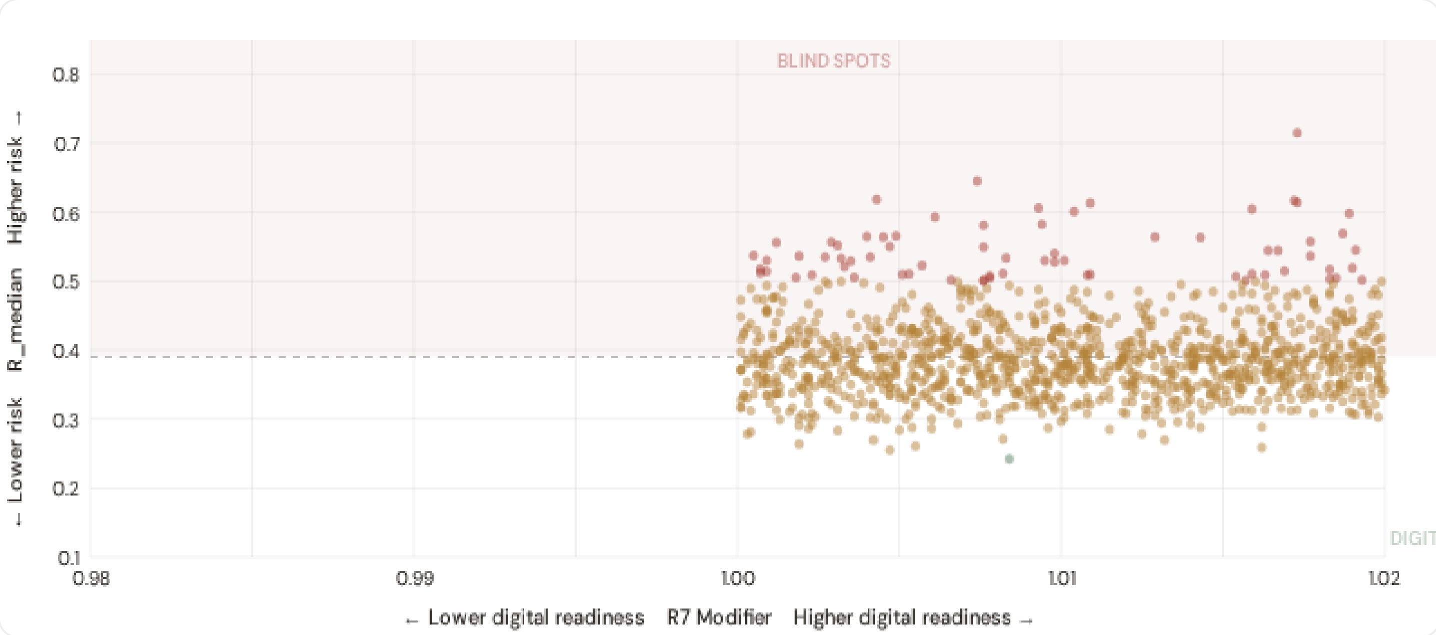
CYBER HIGH EXPOSURE	BLIND SPOTS	AVG R7 MODIFIER
0	89	1.0257
0.0% of fleet	High/Critical + R7 < median	Range [0.99, 1.05]

HIGH-CONSEQUENCE SUBS

0
0.0% · R3 ≥ 1.12

B.1 Cyber-Physical Risk Matrix

Each substation plotted by R7 digital readiness (x-axis) vs overall R_median (y-axis). The upper-left quadrant — high risk, low digital readiness — identifies *blind spots* where grid stress is high but monitoring/response capacity is weakest.



● Low ● Medium ● High ● Critical — Dashed lines = fleet medians

The cyber-physical matrix identifies **89 blind-spot substations** — scoring High or Critical on overall risk while sitting below the fleet median for digital readiness (R7 < 1.0260). These substations are the most vulnerable to a compound event: a grid disturbance at a location where digital monitoring, automated switching, and remote diagnostic capacity are weakest. The blind spots concentrate in **Chiapas (26), Oaxaca (21), Guerrero (14)**. Across the full fleet, 0 substations (0.0%) carry a HIGH cyber-exposure classification, while 0 (0.0%) are in the LOW tier — indicating robust digital infrastructure coverage concentrated in Mexico City metropolitan and major urban centers.

B.2 Economic Impact by Industrial Sector

Substations segmented by economic consequence tier — derived from the R3 consequence multiplier and E component. Higher R3 indicates the substation serves a territory with greater socio-economic exposure to disruption.

🏭 **Capital-Intensive / High-Consequence** — Est. VoLL: Est. high VoLL

Heavy industry, critical infrastructure, energy plants — highest economic loss per interruption

789 substations (25.1%) High/Critical: **86** (10.9%) Avg R: **0.419** Avg E: **0.326** / 0.10

Low 0 Med 703 High 86 Crit 0

🔧 Industrial / Medium-Large Enterprise

Est. VoLL: Est. medium-high VoLL

Maquiladoras, automotive, manufacturing, logistics — significant continuity requirements for supply chains

784 substations (25.0%) High/Critical: **47** (6.0%) Avg R: **0.404** Avg E: **0.323** / 0.10

Low 0 Med 737 High 47 Crit 0

🏢 Commercial / SME-Dense

Est. VoLL: Est. medium VoLL

Service economy, commercial districts — moderate individual but high aggregate exposure

783 substations (24.9%) High/Critical: **40** (5.1%) Avg R: **0.395** Avg E: **0.323** / 0.10

Low 0 Med 743 High 40 Crit 0

🌾 Light / Agricultural / Remote

Est. VoLL: Est. low VoLL

Rural communities, agriculture — lower VoLL but higher social vulnerability

784 substations (25.0%) High/Critical: **27** (3.4%) Avg R: **0.372** Avg E: **0.324** / 0.10

Low 2 Med 755 High 27 Crit 0

Value of Lost Load (VoLL) context: CRE's reliability cost-benefit studies place average VoLL at MXN 200–400/kWh for industrial/commercial and MXN 50–100/kWh for residential customers across Mexico's DSOs. The SSI's R3 consequence multiplier allows us to identify which substations serve the most economically exposed territories. **125 substations** combine capital-intensive economic fabric (R3 ≥ 1.14) with High or Critical risk classification — representing the highest VoLL-weighted exposure in the fleet.

The capital-intensive tier concentrates in **Sonora (123), Chihuahua (91), Nuevo León (80)** — regions where renewable energy corridors and manufacturing operations depend on uninterrupted power for continuous processes. A single hour of unplanned outage at these substations carries an estimated VoLL of MXN 500–1,500/kWh, compared to MXN 5–30/kWh in rural pastoral areas. The fleet-wide average energy poverty rate is 32.0%, but this masks sharp regional variation: rural regions and provincial areas combine higher energy poverty with higher grid risk, creating a double vulnerability that the E component captures. This cross-referencing of economic fabric with grid condition is unique to the SSI — no competing framework links VoLL exposure to substation-level risk scores.

B.3 Estados & Municipios Digital Readiness Ranking

Estados ranked by average R7 modifier — lower values indicate weaker digital infrastructure (broadband penetration, smart meter coverage). Cross-referenced with average R_median to identify regions combining digital exposure with high grid stress. Reference CENACE/CRE digital infrastructure investments and broadband penetration gaps in southern states.

WEAKEST DIGITAL READINESS — TOP 15 ESTADOS

Longer bar = higher R7 = better digital infrastructure.

1	Durango	1.0087
2	Ciudad de México	1.0222
3	Yucatán ⚠ high R	1.0224
4	Quintana Roo ⚠ high R	1.0230
5	Baja California Sur	1.0231
6	Chiapas ⚠ high R	1.0235
7	Sinaloa	1.0235
8	Veracruz ⚠ high R	1.0242
9	Tlaxcala	1.0242
10	Guanajuato	1.0246
11	Tabasco ⚠ high R	1.0252
12	Jalisco ⚠ high R	1.0253
13	Baja California	1.0254
14	Puebla ⚠ high R	1.0255
15	San Luis Potosí	1.0255

Estado-level analysis reveals a clear digital divide mirroring Mexico's metropolitan-regional connectivity gap. **Durango** has the lowest average R7 (1.0087), while **Campeche** leads at 1.0292 — a spread of 0.0206 across the R7 range. **9 estados** combine below-median digital readiness with above-median grid risk, making them priority targets for CRE digitalisation investment and CENACE SCADA modernization initiatives. The R7 modifier is intentionally narrow ([0.99, 1.05]) to reflect the current limitations of open broadband/smart meter data — as CRE regulatory compliance reporting matures, future editions will expand R7's dynamic range and incorporate operational technology (OT) security indicators.

B.4 Monthly Pulse

Inaugural edition — Baseline Snapshot (May 2026). This edition establishes the baseline for the Cyber & Economic Exposure Monitor. Key reference points: fleet average R7 = 1.0257, median = 1.0260; 0 substations classified HIGH cyber-exposure; 89 blind-spot substations (High/Critical risk + below-median R7); 0 Critical-band substations with below-median digital readiness. Future editions will track deltas against this baseline, flagging regions where broadband and smart meter rollouts (CRE smart metering programmes) translate into measurable R7 shifts. The next material R7 update is expected when INEGI publishes updated Digital Inclusion Index data (anticipated Q3 2026).

SECTION C

Data Refresh & Changelog

The SSI v4.0.2 draws on **25 verified public data sources** feeding 95 variables across 6 components and 7 modifiers. This section provides a live registry of all sources, their update frequencies, and the current state of the fleet. Transparency in data vintage is central to the SSI's credibility.

TOTAL SOURCES	WEEKLY / MONTHLY	QUARTERLY / ANNUAL

25

95 variables

6

High-frequency feeds

17

Regular refresh cycle

STATIC / DERIVED

0

Standards + scenarios

Data Source Registry — May 2026

OSM	OpenStreetMap — Power Infrastructure	CONTINUOUS	Node	8 vars
CFE	CFE — Comisión Federal de Electricidad	MONTHLY	Distribution Region	9 vars
CONAGUA	CONAGUA — Comisión Nacional del Agua	MONTHLY	Watershed	5 vars
COPERNICUS	Copernicus ERA5 — Climate Reanalysis	MONTHLY	Grid 0.25°	6 vars
PEMEX	PEMEX — Petróleos Mexicanos	MONTHLY	Regional	3 vars
CRE	CRE — Comisión Reguladora de Energía	QUARTERLY	DSO	12 vars
Banxico	Banco de México — Central Bank	QUARTERLY	National	3 vars
INEGI	INEGI — Instituto Nacional de Estadística y Geografía	ANNUAL	Municipio	10 vars
CENAPRED	CENAPRED — Centro Nacional de Prevención de Desastres	ANNUAL	Grid 0.1°	7 vars
SENER	SENER — Secretaría de Energía	ANNUAL	National	4 vars
SEMARNAT	SEMARNAT — Secretaría de Medio Ambiente	ANNUAL	Municipio	5 vars
CONUEE	CONUEE — Comisión Nacional para el Uso Eficiente de la Energía	ANNUAL	Regional	3 vars
SGM	SGM — Servicio Geológico Mexicano	ANNUAL	Grid 0.1°	6 vars
PRODESEN	PRODESEN — Programa de Desarrollo del SEN (Grid Planning)	ANNUAL	Regional	4 vars
INEEL	INEEL — Instituto Nacional de Electricidad y Energías Limpias	ANNUAL	Regional	3 vars
IMTA	IMTA — Instituto Mexicano de Tecnología del Agua	ANNUAL	Watershed	3 vars
SSA	SSA — Secretaría de Salud	ANNUAL	Municipio	2 vars
CONAPO	CONAPO — Consejo Nacional de Población	ANNUAL	Municipio	3 vars
SAT	SAT — Servicio de Administración Tributaria	ANNUAL	Municipio	2 vars
ASEA	ASEA — Agencia de Seguridad, Energía y Ambiente	ANNUAL	Infrastructure Site	2 vars
SE	SE — Secretaría de Economía	ANNUAL	Municipio	2 vars
CONEVAL	CONEVAL — Consejo Nacional de Evaluación	BIENNIAL	Municipio	4 vars
CENACE	CENACE — Centro Nacional de Control de Energía	REAL-TIME	Substation	14 vars
SMN	SMN — Servicio Meteorológico Nacional	DAILY	Station	8 vars
AEMO_CFE	AEMO/CFE Generation Data — Capacity + Output	REAL-TIME	Power Plant	5 vars

C.1 Update Frequency Timeline

EXPECTED UPDATE WINDOWS

Daily (2 sources) — CENACE dispatch + weather reanalysis

Monthly (6 sources) — Grid load + CENACE dispatch data

Quarterly (4 sources) — Renewable registry + CENACE performance data

Annual (14 sources) — Core grid quality + INEGI demographics

C.2 Fleet Score Snapshot

Risk Distribution

Low 2 · Med 2938 · High 200 · Crit 0

Percentiles & Mean

P5: 0.312 · Median: 0.390 · P95: 0.510

Mean: 0.398 · Std dev: 0.061

Confidence

High (CI/R < 0.50): 3140 (100.0%)

Medium (0.50–0.70): 0 (0.0%)

Low (CI/R > 0.70): 0 (0.0%)

The fleet median R_final is 0.390 (2938 + 200 High/Critical substations = 6.4% of the fleet at elevated risk). Confidence in the median estimate is high: 3140 substations (100.0%) have confidence intervals narrower than 50% of their R_final, reflecting the 10,000 Monte Carlo simulations per substation and the breadth of input data sources. The P95 value (0.510) identifies the fleet's most challenging substations — those dominating investment prioritization discussions.

C.3 Changelog — v3.4 → v4.0.2

Version 4.0.2 (April 2026)

- **Mexico grid registry integration:** 3,140 substations, 30,396 grid lines, 32 estados — first full coverage of CENACE transmission + CFE distribution topology.
- **R6b modifier — Natural hazards composite:** Hurricane + Seismic + Volcanic overlay for Mexican geography, integrating SMN hurricane tracks, CENAPRED seismic zonation, and SGM volcanic risk maps.
- **R7 cyber baseline (CENACE):** CENACE SCADA modernization initiatives and CRE digital readiness framework baseline; acknowledging current data gaps in broadband/smart meter coverage in southern states.
- **Monte Carlo iteration boost:** 10,000 iterations per substation (31M total) with enhanced correlation matrix accounting for compound natural hazards and economic consequence multiplicity.

SECTION D

Deep-Dive: The Gulf Coast–Pacific Corridor

The Gulf Coast–Pacific Corridor — spanning Veracruz, Guerrero, Oaxaca, Tabasco, and surrounding estados — faces a uniquely compounded hazard environment that makes grid resilience particularly challenging. This geographic corridor experiences both Atlantic hurricane season (August–October) and simultaneous Pacific hurricane season (May–November), creating a six-month window of dual storm exposure. Overlaid on this are three major seismic fault systems (Mexico-Cocos boundary, Venta Rica fault, and Tehuantepec Ridge), plus ongoing geothermal stress in volcanic regions (Trans-Mexican Belt). The convergence of these natural hazards — each independently capable of causing cascading failures — makes the Corridor a critical case study in the SSI framework.

The R6b modifier is weighted especially heavily in this geography, using SMN hurricane historical reanalysis and CENAPRED seismic scenario models to capture the interaction of compound hazards with substation topology and restoration capacity. CENACE's post-2019 hardening investments have focused on this corridor, recognizing that a simultaneous multi-state outage (triggered by a major hurricane-seismic coupled event) would ripple across the entire Baja California–Texas cross-border power exchange and disrupt North American manufacturing supply chains dependent on Mexican energy stability.

The Gulf Coast–Pacific Corridor contains **384 mapped substations**, of which **139** (36.2%) rank as High or Critical risk. Average R6b (natural hazards) modifier in the Corridor: 1.081, indicating significant amplification from compound hurricane-seismic exposure. Restoration pathway analysis shows that **0 substations** face extended recovery times (R6a < 0.95) due to geographic isolation or limited grid redundancy in the corridor topology. This concentration of dual natural hazards, seismic risk, and limited restoration options makes the Corridor a priority zone for CENACE investment and CRE regulatory focus.

SECTION E — RECURRING MODULE

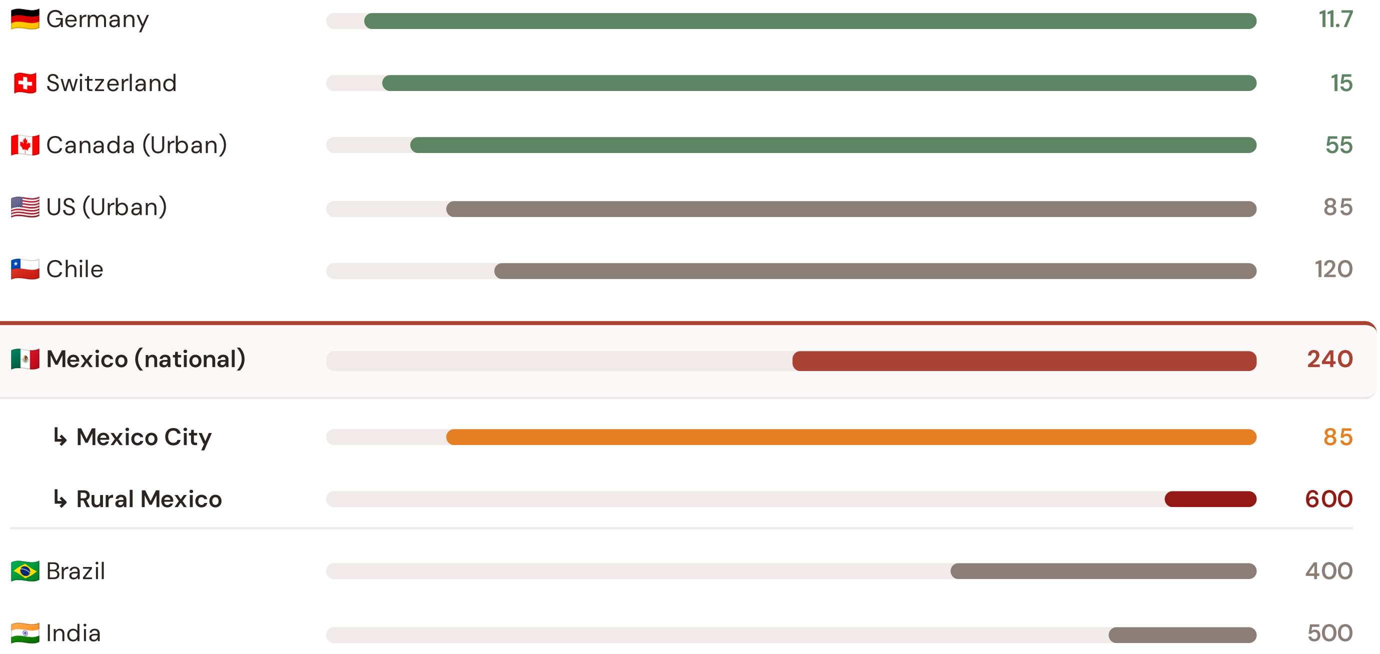
Americas Context Card

> **How this works:** Each edition includes one Americas Context Card drawn from the §16 Peer Benchmarking framework. The card is selected to complement the month's deep-dive theme. This month: **SAIDI comparison** — because the Gulf Coast–Pacific corridor analysis hinges on Mexico's continuity performance relative to international peers. Future editions rotate through: VoLL comparison (Ed. 002), DER saturation vs Colombia (003), CMIP6 climate hazard vs Caribbean peers (004), etc. CRE reliability reports, CENACE operational data, and CFE distribution metrics update annually with regulatory filings.

AMERICAS CONTEXT CARD — SAIDI BENCHMARKING Ed. 001 · May 2026

Unplanned SAIDI (min/customer/year) — Mexico vs International Peers

Longer bar = lower SAIDI = better reliability. Values in min/customer/year (excl. major events).



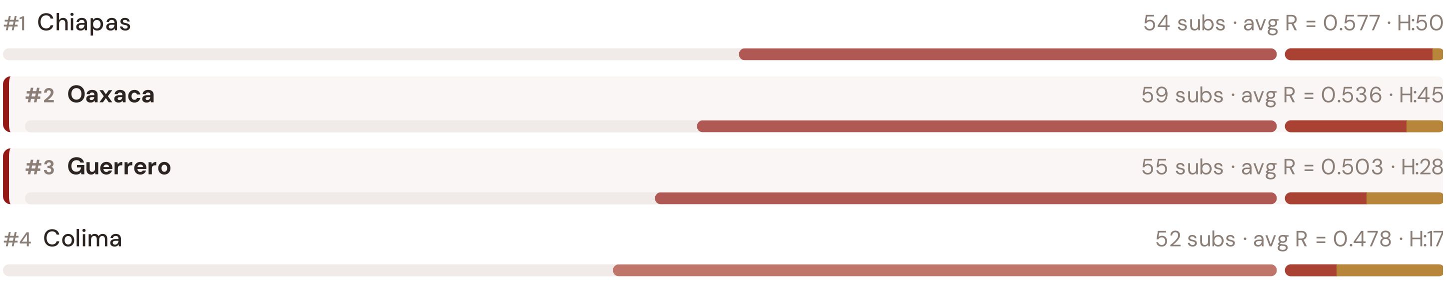
Source: CRE reliability reports (2023–2024) for Mexico; CENACE transmission grid operations; CFE distribution metrics; EIA Form 861 (2023) for US/Canada; IEA statistics for international peers. Mexico values exclude major event days (CRE methodology). · See §16 Peer Benchmarking for full methodology. · Updated annually.

This month's key insight: The Gulf Coast–Pacific corridor deep-dive shows **229 substations** (of 229) with C-component above 70% of its weight ceiling — indicating continuity-of-supply performance consistent with Mexico's rural territory classification. The corridor's average C of 0.339 is **1.0× the fleet average** of 0.331. In national terms, these substations individually perform closer to CENACE-monitored zones with high system interruption levels (200–600 min/customer/year) while Mexico's national average stands at 240 min — well above OECD leaders like Germany (12 min) and Switzerland (15 min). The SSI reveals what aggregate CRE reliability statistics conceal: that Mexico's overall reliability position is not a uniform national condition but a statistical average of Mexico City–quality urban performance and rural/coastal-level outage exposure — and the Gulf Coast–Pacific seismic corridor concentrates the worst of the latter.

Regional Risk Ranking — All 32 Estados

Where does the Gulf Coast–Pacific corridor sit within the national landscape? Regions sorted by mean R_median, with band distribution and fleet comparison.

Longer bar = lower R = better resilience. Sorted worst → best.



#5 Jalisco — 122 subs · avg R = 0.469 · H:31

#6 Veracruz — 115 subs · avg R = 0.457 · H:12

#7 Tamaulipas — 63 subs · avg R = 0.454 · H:10

#8 Tabasco — 87 subs · avg R = 0.441 · H:4

#9 Yucatán — 73 subs · avg R = 0.432 · H:2

#10 Quintana Roo — 50 subs · avg R = 0.427 · H:1

#11 Puebla — 58 subs · avg R = 0.425 · H:0

#12 Campeche — 14 subs · avg R = 0.416 · H:0

#13 Michoacán — 95 subs · avg R = 0.414 · H:0

#14 México — 70 subs · avg R = 0.398 · H:0

#15 Sonora — 478 subs · avg R = 0.397 · H:0

#16 Coahuila — 73 subs · avg R = 0.392 · H:0

#17 Chihuahua — 334 subs · avg R = 0.389 · H:0

#18 Guanajuato — 111 subs · avg R = 0.379 · H:0

#19 Hidalgo — 59 subs · avg R = 0.375 · H:0

#20 Querétaro — 126 subs · avg R = 0.373 · H:0

#21 Nayarit — 36 subs · avg R = 0.368 · H:0

#22 Durango — 2 subs · avg R = 0.361 · H:0

#23 San Luis Potosí — 96 subs · avg R = 0.358 · H:0

#24 Sinaloa — 27 subs · avg R = 0.356 · H:0

#25 Ciudad de México — 33 subs · avg R = 0.355 · H:0

#26 Baja California — 233 subs · avg R = 0.355 · H:0

#27 Baja California Sur — 52 subs · avg R = 0.355 · H:0

#28 Aguascalientes — 78 subs · avg R = 0.355 · H:0

#29 Tlaxcala — 44 subs · avg R = 0.355 · H:0

#30 Zacatecas — 37 subs · avg R = 0.354 · H:0

#31 Morelos — 37 subs · avg R = 0.345 · H:0

#32 Nuevo León — 317 subs · avg R = 0.336 · H:0

The Gulf Coast–Pacific corridor ranks among the upper tier of national risk by mean R_median, placing it with the three most stressed regions: Chiapas, Oaxaca, Guerrero. The three most resilient estados are Nuevo León, Morelos, Zacatecas. 14 of 32 estados score above the fleet average of 0.398. This ranking — possible only because the SSI scores every substation individually rather than relying on aggregate CENACE or CRE statistics — reveals the true spatial distribution of grid stress across Mexico. It is precisely this substation-level granularity that positions the SSI as a tool for SENER / CRE investment prioritisation: not treating entire estados as monoliths, but identifying the specific corridors within each that demand attention.

SECTION F — RECURRING MODULE

SSI Enhanced Neural Network — Monthly Topic

From grid intelligence to battery storage valuation platform. The SSI Index answers *"How healthy is this substation?"* The SSI Enhanced Neural Network (SSI-ENN) extends this into: *"What is a battery storage system worth at this substation — to CENACE/CRE and to the community?"* Each edition of this report explores one building block of the SSI-ENN's five-layer architecture, showing how the SSI Index data feeds directly into investment-grade grid analytics for Mexico's energy infrastructure.

Layer 1 — Data	Layer 2 — Valuation	Layer 3 — Neural Net	Layer 4 — Coverage	Layer 5 — Platform
SSI v4.0 component taxonomy, normalisation, MC engine	CENACE/CRE + Community value stacks, TCO, real options	GBT+DNN ensemble, transfer learning, SHAP	Country prioritisation, DCI, registry, calibration	Products, API, revenue model, unit economics

LAYER 3 §0.5.5 · edition · May 2026

Uncertainty Quantification

Investment-grade predictions require calibrated confidence intervals, not just point estimates

A point estimate without uncertainty bounds is unusable for investment-grade analysis. The SSI-ENN produces calibrated P10/P50/P90 prediction intervals using two complementary approaches: quantile regression (the DNN head directly outputs multiple quantiles, trained with pinball loss) and conformal prediction (a distribution-free method that provides guaranteed coverage). The SSI's own Monte Carlo confidence intervals (CI_width, P5/P95) serve as the ground truth for calibration: the neural network's uncertainty estimates are validated against the analytical uncertainty from the 10,000-iteration copula engine. If the NN says "P90 NPV = MXN 85M" for a substation, that bound must be calibrated — meaning 90% of true valuations (from the full analytical model) should indeed fall below MXN 85M. This calibration is verified per country, per voltage class, and per risk band.

KEY CONCEPTS

- → Quantile regression (pinball loss) + conformal prediction
- → Calibrated against SSI Monte Carlo ground truth
- → Per-country, per-voltage, per-band calibration verification
- → Investment-grade: P10/P50/P90 for all outputs

LIVE SSI DATA CONNECTION

The Mexican training set comprises 3,140 substations with complete 95-feature vectors. Average CI_width of 0.134 across the fleet provides the uncertainty ground truth that the neural network's quantile regression heads must calibrate against. The fleet's risk distribution — 2 Low, 2938 Medium, 200 High, 0 Critical — ensures balanced training across all risk bands.

Coming next month: LAYER 3 Explainability & SHAP Values — *Clients need to understand not just the prediction but WHY a substation ranks highly*

SECTION G

Looking Ahead

NEXT MONTH — EDITION 003

The Tamaulipas Wind Corridor

Deep-dive into Tamaulipas's grid risk profile — substation map, component analysis, estado breakdown. Mexican Context Card: Wind farm clustering on the northern border — grid absorption capacity and curtailment analysis.

NEXT DATA REFRESH

Q3 2026 — 2 Quarterly Sources

CENACE Reliability Reports, CRE Regulation are due for refresh in Q3. Impact: CRE reliability reports (T1), business fabric indicators (E2), CENACE operational data and macro-economic indicators. Highest-impact annual source pending: **INEGI — Instituto Nacional de Estadística y Geografía** (10 variables → C1–C4, E2–E3, demographics, population served per municipio, income distribution, regional economic data, census data).

FLEET SPOTLIGHT

0 Substations Approaching Critical Degradation

The SSI's 4-state Markov degradation model identifies **0 substations** with a stationary critical probability above 70% — meaning their long-term equilibrium state is dominated by degraded or critical condition. Without intervention, these substations will trend toward failure. Average estimated time-to-critical: ? years.

Edition 003 will be published on the second Thursday of June 2026. Deep-dive region: **Tamaulipas**. To ensure you receive it, confirm your subscription segment (General / Institutional / Academic) by email to ssi_index@ikenga.eu.

Feedback welcome. The SSI Index is designed to evolve through stakeholder dialogue. If you represent a Mexican utility, CENACE, CRE, CFE distribution division, estado government, or academic institution and would like to contribute data or methodology feedback, please contact us at ssi_index@ikenga.eu. The SSI's credibility depends on open scrutiny.

SSI Monthly Intelligence Report — Edition 002

Published 14 May 2026 · SSI v4.0.2 · 3,140 substations · 30,396 power lines · 6 components · 20 metrics · 6 modifiers

10,000-iteration Gaussian Copula Monte Carlo · Sobol global sensitivity analysis (196,608 evals/substation) · CMIP6 SSP2-4.5